SahayakSHG: Project Proposal

Tech4Bharat: GAVS Hackathon 2026 | Theme: Youth and Women | Problem Statement 8.1

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How to read this document. Facts taken from public sources carry a tag like [R4] that points to the References section at the end. Our evidence so far is desk research from public sources, and field interviews with SHGs are still planned.

Summary

SahayakSHG is a voice-first, multi-agent AI assistant for Self-Help Group (SHG) leaders. The leader speaks in her own language, and the system turns a buyer's bulk order into a fair price, a fair split of work, a tracked delivery and a transparent payout. Every member can hear her own record read aloud, and the group approves every work split and every payout. Over time each group builds a delivery-reliability record that buyers can trust, which turns an informal group into a supplier worth seeking out.

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1. The problem and the evidence

The scale. DAY-NRLM data reported by the Ministry of Rural Development lists 94.16 lakh SHGs with 10.03 crore members, organised into 5.62 lakh village organisations and 34,314 cluster-level federations [R1]. The government has set a Lakhpati Didi target of six crore women, and 3.46 crore had reached the status by 31 March 2026 [R5].

What the brief says. The problem statement describes a leader dividing a bulk order across 10 to 15 members verbally, tracking it in a notebook, checking quality only when output comes together and splitting payment by estimate. It reports that members underprice by 30 to 40 percent because material, labour and margin are never calculated, and that many members have low literacy.

Bulk orders to SHGs are real and large.

- Telangana ordered school-uniform stitching exclusively from women's SHGs and their federations in March 2024 [R6], with an estimated 63.4 lakh pairs to be stitched and about 28,200 SHG women tailors needed to finish in 45 days [R7].
- Maharashtra assigned about 88 lakh uniforms to SHGs through a state body, at a reported Rs 110 per uniform [R10].
- In Bihar, uniforms stitched by Jeevika SHG women were reported to reach around 50 lakh Anganwadi children [R27].

Where the process strains.

- Madhya Pradesh (2019): an official said SHGs could not provide the quality and quantity needed for a uniform order [R11].
- Maharashtra (2024): an SHG engaged for uniforms failed to deliver on time, the second uniform was still pending half way through the session, and the state moved back to an older scheme [R9].
- Telangana (June 2026): supply was delayed, SHG women were paid Rs 75 per uniform, and an education department source claimed bulk machine stitching would cost Rs 35 to 40 [R8]. This shows both price pressure on groups and why knowing your own unit cost matters.
- These are news reports about system-level programmes, so they show risk and context rather than proving the cause in every group.

When groups are organised, they can deliver. In Godda, Jharkhand, an SHG of about 150 women received a 30 percent advance on a uniform contract, and over three lakh uniforms were reported delivered by the end of September 2019 [R12].

Academic evidence.

- A study of 288 SHG members in Madurai and Coimbatore found inadequate marketing facilities to be the first-ranked problem, with lack of formal education reported by 55.9 percent, internal group conflicts by 53.8 percent and members leaving by 43.75 percent [R13].
- A study of jute handicraft SHG entrepreneurs in Cooch Behar recommends hands-on training in pricing, packaging, branding, e-marketing and e-payment [R14].
- A study of fisherwomen SHGs lists lack of remunerative price among the production and marketing constraints [R15].
- A national SHG enterprise toolkit from the livelihood mission ecosystem stresses structured market surveys, pricing and role allocation inside groups [R17].

What we do not yet have. We have not yet spoken with an SHG leader, and this document does not contain any invented quotes. Section 12 gives the field questions and plan for when contact is possible.

2. What exists today, and the gap

Existing effort	What it does	Gap we address	How we relate
LokOS (DAY-NRLM)	Digitises SHG, VO and CLF registers, savings, loans, repayments and livelihood data [R1]	Records group finance, and is not described as running a single customer order end to end	Complement; export order and payout summaries in a LokOS-friendly format

SHE-LEAPS (June 2026)	Enterprise creation, performance tracking and business management for SHG enterprises; over 11 lakh enterprises reported registered [R2, R3]	Public descriptions stress monitoring and data capture; we found no public description of order-level pricing, member-level allocation or approved payouts	Complement; verify with a demo or livelihood-mission staff
e-Shakti (NABARD)	Digital bookkeeping for SHGs to support bank appraisal [R25]	Focus is savings, loans and credit history, not order fulfilment	Complement
Private SHG bookkeeping apps	Digital records of meetings, savings and loans	Same as above	Complement
Notebook, verbal instructions, WhatsApp	What most groups use for orders today (per the problem statement)	No costing, fairness, approval or audit trail	Replace for the order workflow

Positioning. SahayakSHG is the order-to-payout operations layer inside a group's enterprise: price it, split it, track it, pay it, and prove it. It is voice-first for members with low literacy.

3. Users and stakeholders

Stakeholder	What they need	What SahayakSHG gives them
SHG leader	Quote correctly, divide work fairly, avoid disputes	Pricing suggestion, work split with reasons, deadline alerts, payout matrix
SHG members	Know what they earned and why, and trust the accounts	Audio playback of her own record, visible group approval
Buyers	On-time, consistent-quality delivery	Progress visibility and, later, a reliability record

Federations and livelihood mission staff	See which groups are order-ready, reduce disputes	Group-level dashboard, standard records
Material suppliers and linked banks	Clean, auditable records	Ledger exports (with consent)

4. How the solution works

The worked example below uses made-up numbers to show the logic: a buyer wants 500 school bags in 20 days from a group of 12 members.

1. **Order intake.** The leader speaks the order in her language. The Intake agent transcribes it and extracts product, quantity, deadline and number of members, and asks follow-up questions for anything missing. The leader confirms the parsed order by audio or tap.
2. **Pricing.** The Pricing agent asks for material cost, labour hours and overheads, then plain code computes the price. Example: material 120, labour 1.5 hours at 40 = 60, overhead 10 percent of 180 = 18, cost 198, margin 20 percent gives 237.6, rounded to 238. A quote of 150 would lose money on every bag.
3. **Allocation.** The Allocation agent proposes a split using each member's available time, skill and recent workload, and explains it in simple language. The leader can override, the group approves, and the override is recorded.
4. **Progress and quality.** Members report output by voice or tap. Early sample photos are checked so that problems appear on day 2 and not on day 20. The leader has the final word on quality, because quality in handmade goods is subjective.
5. **Consolidation and risk.** The Tracking agent compares totals with the deadline and flags risk early, for example: one member is behind, move 40 pieces to two others?
6. **Payout.** Earnings are computed from accepted pieces at the approved piece rate. A reconciliation check confirms that member payouts, group fund and material cost equal the buyer's payment. In the example, 500 bags at 238 give 119,000, which equals material 60,000 plus labour 30,000 plus overhead 9,000 plus margin 20,000. The group approves the payout, and each member can hear her own statement.

Human checkpoints. The leader confirms the parsed order. The group approves the work split. The leader decides quality. The group approves every payout. No agent can lock a record or release a payout on its own.

5. Designing for low literacy and low connectivity

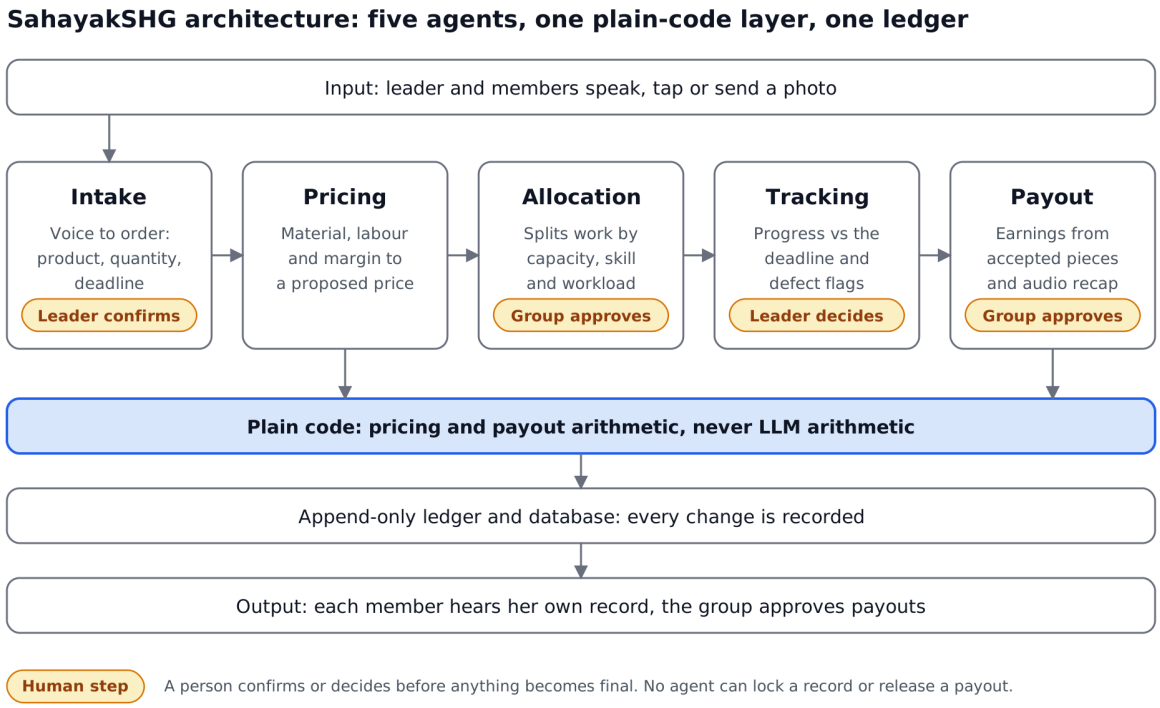
What the research says.

- Work with illiterate and semi-literate domestic workers in Bangalore found that text-free interfaces, built on semi-abstracted graphics and voice feedback, were strongly preferred over text interfaces, and that a consistent help feature mattered [R18].
- A later study of low-literacy users in several countries found textual interfaces unusable for first-time low-literacy users. In the mobile banking test, a graphical interface had the highest task completion, while people who understood the spoken dialogue system used it faster [R19]. The lesson is to combine icons and voice, not to rely on voice alone.
- A field pilot with rural women entrepreneurs reports that those who received frequent hand-holding visits did better with a finance app, and that they noted transactions in a daybook before entering them in the app [R21].

What we do with it.

Principle	Design choice in SahayakSHG
Icons plus voice	Every screen is icon-led, with a play-audio button that reads the screen in the member's language
Numbers must be heard	Pay, pieces and deadlines are read aloud, and the same numbers are shown as large digits
Assisted first	The leader operates the app for the group; members get a simple view-and-listen screen of their own record
Consistent help	The same help button sits in the same place on every screen [R18]
Habits first	The app mirrors a notebook: one order, one list of members, one tally. Photo capture of notebook pages is a later feature
Onboarding through people	DAY-NRLM already has about 4.09 lakh community resource persons [R1]; they are the natural trainers for first use
Poor networks	Updates are saved on the phone and synced when connectivity returns

6. Architecture and technical design



The leader's voice or a member's tap enters through the Intake agent, then the order moves through Pricing, Allocation, Tracking and Payout. Agents understand, ask and explain. They never do the final arithmetic. Every number is calculated in plain code and written to an append-only ledger, the single source of truth for disputes. The group approves the work split and the payout before they become final.

Agents

Agent	Input	Output	Language model used for	Plain code used for
Intake	Voice or typed text in a regional language	Structured order: product, quantity, deadline, members	Understanding speech and asking follow-up questions	Validating numbers and dates
Pricing	Order plus material cost, labour hours, overhead, target margin	Proposed unit price and total, with a plain explanation	Asking for missing inputs and explaining the price	Cost and margin calculation

Allocation	Order plus members' capacity, skill and recent workload	Quantity per member, with reasons	Explaining the split and handling leader overrides	Weighted split with exact rounding
Tracking and quality	Output entries by voice or tap, early sample photos	Progress against deadline, risk alerts, redistribution suggestions	Reading voice updates and summarising risk	Totals, projection, accepted and rejected counts
Payout	Accepted pieces per member and the approved piece rate	Payout matrix, ledger entries, audio statement per member	Writing the spoken statement	All payout arithmetic and the reconciliation check

Calculations in plain code

All numbers below are illustrative.

Pricing. Cost per piece = material + labour hours x hourly rate + overhead. Overhead = overhead percentage x (material + labour). Price = cost x (1 + target margin). On the example in section 4, a quote of 150 against a cost of 198 would lose 48 on each of 500 bags, which is 24,000 in total.

Allocation. Weight of member i = available capacity x skill factor x fairness factor, where the fairness factor rises for members who carried less in recent orders. Share = total quantity x weight / sum of weights. Rounding uses the largest-remainder method so that the shares add up exactly. Example: 100 pieces and three members with weights 1.0, 0.8 and 1.2 give raw shares 33.33, 26.67 and 40.00. The floors are 33, 26 and 40, which add to 99, and the extra piece goes to the member with the largest remainder, giving 33, 27 and 40. The weights and reasons are stored so that the explanation shown to the group matches the real calculation.

Payout. Earning = accepted pieces x piece rate, less any deduction the group has agreed for rejected pieces. The piece rate comes from the labour part of the approved price. Margin is split by a rule the group chooses (equal share, or proportional to pieces). A reconciliation check requires that member payouts plus group fund plus material cost equal the buyer's payment. If not, approval is blocked and the gap is shown.

Data model

Table	Key fields

Group	id, name, leader_id, language
Member	id, group_id, name, skill_level, weekly_capacity_hours, consent_given
Order	id, group_id, product, quantity, deadline, buyer, status
PriceQuote	id, order_id, material_cost, labour_cost, overhead, margin, unit_price, approved_by
Task	id, order_id, member_id, assigned_qty, reason, approved
OutputEntry	id, task_id, qty, quality_status, photo_ref, reported_by, timestamp
Payout	id, order_id, member_id, amount, rule_used, group_approved
LedgerEntry	id, timestamp, actor, action, before, after (append only, never edited)

A group has many members and orders. An order has one price quote, many tasks and many payouts. A task has many output entries. Every change to any table writes a ledger entry.

Stack and platform choices

Layer	Primary choice	Fallback
Frontend	Mobile-first web app that installs like an app, in React or Next.js	Plain responsive web app
Backend	FastAPI (Python) holding the calculation code	Node with Express
Database	SQLite for the prototype, Postgres for a demo deployment	SQLite throughout

Speech to text and text to speech	Bhashini [R22]	Another Indian-language speech service, or browser speech recognition for the demo
Agent orchestration	Lyzr [R23]	Direct language-model calls with our own orchestration code
Offline	Local queue in the browser, sync on reconnect	Show queued status only

Lyzr. Its public documentation describes managerial orchestration for dynamic workflows, DAG-style workflows for fixed ones, custom tools built from OpenAPI or code, and REST APIs [R23]. That fits our design: the pricing and payout functions are exposed to the agents as tools, so the model calls the code and never replaces it. Details such as hackathon credit limits are still to be confirmed with the organisers.

Bhashini. A national platform that offers speech recognition, text to speech and translation across 22 Indian languages through APIs, with registration to obtain credentials [R22]. Accuracy on regional dialects and noisy rooms is untested for our use and is listed as a risk.

7. Trust, fairness and privacy

- **Every member sees and hears her own record.** The group approves every payout before it is final.
- **Append-only ledger.** Changes are added, never overwritten, so a dispute can be settled from the shared history.
- **Explainable allocation.** The reasons for each member's share are shown and stored, and a leader override is recorded.
- **No automated credit scoring in the first version.** The reliability record is built only from order facts (on time, accepted pieces) and is shared with buyers only with the group's consent.
- **Consent and data protection.** India's Digital Personal Data Protection Rules, notified in November 2025, require a clear, itemised, plain-language notice and an easy way to withdraw consent [R24]. Reported timelines phase the main obligations in over about 18 months, to around May 2027 [R24]. We design for this from the start: a short spoken consent in the member's language, minimal data, no storage of identity numbers in the first version, and each member sees only her own record.
- **Humans decide.** No agent can lock a record or release a payout.

8. Venture model

Who pays. State livelihood missions, SHG federations and NGOs pay a small per-group subscription for the order workflow. Buyers and marketplaces can later pay a commission for orders routed to groups with a proven record.

Why groups stay. The tool pays for itself in avoided losses. In the section 4 example, a mispriced quote loses 24,000 on one order.

Rollout. Pilot with about 10 groups under one federation, then extend to a block or district through the federation structure, then replicate across states. An average cluster-level federation covers roughly 270 SHGs (our calculation from 94.16 lakh SHGs and 34,314 federations [R1]). Livelihood-mission community resource persons act as trainers [R1].

Illustrative economics (to be validated). At a hypothetical subscription of Rs 50 per group per month, a federation with 274 groups fully adopting would generate about Rs 1.64 lakh a year. The real figure depends on adoption and on what missions are willing to pay, so this must be tested in a pilot.

First use case. Government bulk programmes such as school and Anganwadi uniforms are a natural entry point, because the order sizes, deadlines and per-piece rates are public [R6, R7, R8, R10].

Metrics for a pilot. Share of orders delivered on time, margin achieved against the group's earlier quotes, time taken to prepare a quote, member retention, and number of payout disputes.

Competitive risk and response. Government platforms such as SHE-LEAPS may add order-level features [R2]. Our response is to integrate rather than compete, and to stay the voice-first, member-facing order layer.

9. Risks and mitigations

Risk	Mitigation
Speech recognition struggles with dialects and noise	Confirm the parsed order back in audio, allow tap correction, test several services
The language model produces a wrong number	All numbers are computed in code, and the model only explains them
Members cannot use a phone	The leader enters updates for them, and members listen to records
Poor connectivity	Local queue, sync later, small payloads

Quality is subjective	The leader has the final word, and early sample photos catch problems sooner
Pay disputes	Append-only ledger, group approval and a reconciliation check
Allocation seen as unfair	Reasons shown and stored, overrides recorded
Adoption	Assisted onboarding by community resource persons [R1, R21]
Government platforms add similar features	Integrate and complement [R2]
Privacy	Consent in the member's language, minimal data, each member sees only her own record [R24]
Tool dependency	Keep an agent interface that can swap Lyzr for direct model calls

10. Validation plan

When contact with SHGs becomes possible (through family networks, a block-level livelihood mission office or a federation), the team will speak with at least two SHG leaders, one or two members, one community resource person and, if possible, one buyer. Ask for permission before using any quote.

Questions drawn from the problem statement.

1. How was the last bulk order divided, and who decided?
2. How do members know what they earned?
3. What happened when a member fell behind?
4. How is quality checked, and who has the final word?
5. How was the price for the order worked out?
6. How do members who cannot read keep track of their work today?

Hypotheses to test. Groups quote without a costing sheet. Members trust audio more than a printed sheet. Leaders will use a voice tool if a community resource person helps at first. Federations see value in a standard order record.

References

ID	Source	Link	Used for
R1	PIB factsheet: LokOS, A Digital Backbone for Rural Livelihoods (4 Jul 2026)	https://pib.gov.in/FactsheetDetails.aspx?Id=150688	SHG and federation counts, LokOS features, community resource persons
R2	PIB factsheet: SHE-LEAPS, Digital Empowerment for Lakhpati Didis (1 Jul 2026)	https://pib.gov.in/FactsheetDetails.aspx?Id=150655 and PDF https://static.pib.gov.in/WriteReadData/specificdocs/documents/2026/jul/doc202671907501.pdf	What SHE-LEAPS is and its objectives
R3	IMPRI: SHE-LEAPS 2026 policy update	https://www.impriindia.com/?p=75942	Reported count of registered SHG enterprises (secondary source)
R4	Lok Sabha answer on DAY-NRLM (Sansad)	https://eparlib.sansad.in/bitstream/123456789/2996949/1/AU374_upreD2.pdf	Official parliamentary figures on SHGs
R5	Kashmir Life: Lakhpati Didis, Lok Sabha told	https://kashmirlife.net/over-2-52-lakh-women-in-jammu-kashmir-become-lakhpati-didis-lok-sabha-told-445266/	Lakhpati Didi progress and target (secondary source)

R6	Telangana Today: uniform orders to women SHGs (12 Mar 2024)	https://telanganatoday.com/telangana-govt-issues-school-uniform-orders-to-women-shgs	Bulk orders awarded to SHGs
R7	Deccan Chronicle: SHGs to stitch uniforms	https://www.deccanchronicle.com/news/shgs-to-stitch-uniforms-for-school-kids-885071	Order size and tailors needed
R8	Telangana Today: uniform supply likely delay (Jun 2026)	https://telanganatoday.com/uniform-supply-likely-delay-to-hit-telangana-government-schools	Delays and per-piece rate
R9	The Hitavada: free uniform policy (21 Dec 2024)	https://thehitavada.com//Encyc/2024/12/21/Free-uniform-to-Govt-school-students-from-next-session.html	SHG delivery delay in Maharashtra
R10	Free Press Journal: Maharashtra uniforms via MAVIM	https://www.freepressjournal.in/education/maharashtra-students-may-not-get-their-uniforms-on-school-reopening-day-at-msps-school	Order size and rate per uniform
R11	Hindustan Times (13 Feb 2019): no uniform for MP schoolkids	https://www.pressreader.com/india/hindustan-times-lucknow/20190213/281986083817107	Quality and quantity gap
R12	The Leaflet: Godda SHG uniform story	https://theleaflet.in/?p=30045	Positive case with an advance payment
R13	Karuna Jeba Mary and Murugan: constraints faced by SHG members (PDF)	https://www.phytojournal.com/archives/2019/vol8issue2S/PartS/SP-8-2-185-417.pdf	Survey of 288 SHG members

R14	Tarafdar: marketing difficulties of jute handicraft products, SHG entrepreneurs (PDF)	https://ir.nbu.ac.in/bitstreams/db3175cc-f088-4334-bed8-4f298095ea60/download	Pricing and marketing training need
R15	Jayaraman: production and marketing constraints of fisherwomen SHGs (PDF)	https://epubs.icar.org.in/index.php/IJF/article/download/41896/29380/486092	Lack of remunerative price
R16	Chakraborty and Chaturvedi: marketing problems of SHGs	https://econpapers.repec.org/RePEc:jct:journl:v:15:y:2020:i:2:p:30-35	Background on SHG marketing problems
R17	Ready reckoner for sustainable enterprise for SHGs (PDF)	https://umcasia.org/wp-content/uploads/Ready-reckoner-for-Sustainable-Enterprise-for-SHGs_English.pdf	Enterprise planning, pricing and roles in groups
R18	Medhi, Sagar and Toyama: Text-free user interfaces (PDF)	https://courses.cs.washington.edu/courses/cse490d/09au/papers/Text-FreeUI.pdf	Graphics plus voice for non-literate users
R19	Medhi Thies and others: mobile interfaces for low-literacy users (ToCHI) and CHI 2009 paper (PDF)	https://www.microsoft.com/en-us/research/?p=161317 and https://microsoft.com/en-us/research/wp-content/uploads/2016/02/medhi_chi2009.pdf	Text versus graphical versus spoken interfaces

R20	IFMR LEAD: Enabling digital adoption among rural women entrepreneurs, baseline (PDF)	https://ifmrlead.org/wp-content/uploads/2023/09/Enabling-Digital-Adoption_STREE-Baseline_July-2023.pdf	Digital adoption context
R21	IFMR LEAD: E-nabling women entrepreneurs (PDF)	https://ifmrlead.org/wp-content/uploads/2023/12/E-nabling-Women-Entrepreneurs_Report_December-2023.pdf	Hand-holding and daybook-first practice
R22	Bhashini overview (Anantam IAS) and Apolitical case study	https://anantamias.com/bhashini-bhasha-interface-for-india/ and https://apolitical.co/en/navigator/case-studies/india-using-ai-to-make-government-services-available-in-22-languages	Speech, text to speech and translation across 22 languages (secondary sources)
R23	Lyzr documentation	https://docs.lyzr.ai/introduction/getting-started/intro.md and https://docs.lyzr.ai/enterprise/get-started/intro.md	Orchestration, custom tools, APIs
R24	Digital Personal Data Protection Rules 2025: Khaitan and Co briefing (PDF), PIB backgrounder mirror, TaxGuru timeline summary	https://www.khaitanco.com/sites/default/files/2025-11/ERGO%20-%20Digital%20Personal%20%20Data%20Protection%20Rules%20-%2015%20November%202025.pdf and https://dpdp.myndsolution.com/wiki/sources/files/pib-2190655.html and https://taxguru.in/?p=1064181	Consent notice duties and phased timeline
R25	NABARD e-Shakti (DSIJ report)	https://insights.dsij.in/article-details/articleid/22025/nabard-initiates-e-shakti-to-help-shgs	Earlier SHG digitisation effort

R26	Digital slates and pens for microfinance in SHGs (PDF)	https://sunchak.pages.iu.edu/publications/92-dig-slates-microfinance-ictd2010-final.pdf	Earlier ICT work with SHGs
R27	Patna Press: Jeevika uniforms for Anganwadi children	https://patnapress.com/?p=57115	Scale of SHG uniform work in Bihar (secondary source)